



Mark Scheme (Results)

June 2023

Pearson BTEC Nationals
In Applied Science/Forensic and Criminal
Investigation (31617H/1C)
Unit 1: Principles and Applications of Science I
Chemistry

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Unit 1: Principles and Applications of Science I Chemistry

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Mark
1(a)	C same number of occupied electron shells	1

Question Number	Answer	Additional Guidance	Mark
1(b)	Any one from: Si, P, S, Cl	allow silicon, phosphorous, sulfur, chlorine. do not allow chloride ignore Ar / argon	1

Question Number	Answer	Additional Guidance	Mark
1(c)	Na (1) Mg (1) Can be in either order	allow sodium allow magnesium	2

Question Number	Answer	Additional Guidance	Mark
1(d)	Cl / Cl ₂	allow chlorine do not allow chloride / Cl ⁻	1

Question Number	Answer	Additional Guidance	Mark
1(e)	an explanation linking four from: melting point increases (from Na to Al) (1) more electrons (lost from outer shell / in delocalised sea of electrons) (1) ions are more positively charged / have more protons / greater (effective) nuclear charge (1) same amount of shielding (1) stronger (electrostatic) attraction between {(positive) ions / nuclei} and delocalised electrons / stronger metallic bonding (1)	allow ORA throughout allow same number of shells reject intermolecular forces	4

	larger amount of energy to break {the (electrostatic) attraction/ (metallic) bonding} (1)	reject intermolecular forces	
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Question Number	Answer	Mark
2(a)	(2) OR δ+ on H and δ- on N of each atom in H bond (1) hydrogen bond (1)	2

Question Number	Answer	Mark
2(b)(i)	C (NH ₄) ₂ SO ₄	1

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	shared pair of electrons (1) (both / two / pair of) electrons come from one {atom / element} (1)	allow dot and cross diagram where both electrons are either dots or crosses to show dative covalent bond for both marks	2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	calculation of number of moles of both $\frac{500}{17} = 29.41$ and $\frac{1000}{63} = 15.87$ (1) limiting reagent is the nitric acid (1)	MP2 is dependent on MP1 but allow ECF	2

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	conversion of volume (1) $\frac{50}{1000} = 0.05$ calculation of number of moles (1) $0.05 \times 0.3 = 0.015$ calculation of mass (1) $0.015 \times 17 = 0.255$	0.255 / 0.26 with or without working scores 3 marks allow ECF throughout allow rounding of final answer if no other marks allow 1 mark for 0.05 seen	3

Question Number	Answer	Mark
3(a)	A decreases across the period	1

Question Number	Answer	Additional Guidance	Mark
3(b)	$1s^2 2s^2 2p^6$	ignore any charges reject subscripts	1

Question Number	Answer	Additional Guidance	Mark
3(c)	an explanation linking any three from: ionic radius decreases (1) the positive ions have the same number of electrons (1) the number of protons is increasing / nuclear charge is increasing (1) there is greater (electrostatic) attraction between electrons and {protons / nucleus} (1)	allow ORA throughout allow there are the same number of shells / same amount of shielding	3

Question Number	Indicative content	
4	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content but learners should be rewarded for other relevant answers. • 14 points on graph so therefore 14 electrons so therefore there are 14 protons / atomic number 14 • 1st section has 4 points, which is 4 electrons in outermost shell • lowest ionisation energy so furthest from the nucleus / most shielded because it is the outermost shell • so silicon is in group 4 • 3 sections / 2 jumps in the graph so therefore 3 shells of electrons • so silicon is in period 3 • points in the same section of the graph means electrons have similar energy • 8 points in the middle section means that there are 8 electrons in second shell • 2 points in the highest section means that there are 2 electrons in first / inner shell • electron configuration is 2.8.4 • gradual increase in each section of the graph as nuclear charge to electron ratio increasing • electrons in same shell • with the same amount of shielding • large jump on the graph / large increase in ionisation energy as the next electron removed is in a new shell • which is closer to the nucleus / less shielding / more electrostatic attraction to overcome Accept any other valid response.	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Adequate interpretation, analysis and/or evaluation of the scientific information with generalised comments being made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Good analysis, interpretation and/or evaluation of the scientific information. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Comprehensive analysis, interpretation and/or evaluation of all pieces of scientific information. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear, coherent and logical.



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